2015-2016 – Econ590
Topics in Macroeconomics

Liquidations and Deficient Demands : Reconciling Hayek's and Keynes' views of recessions - Part 2

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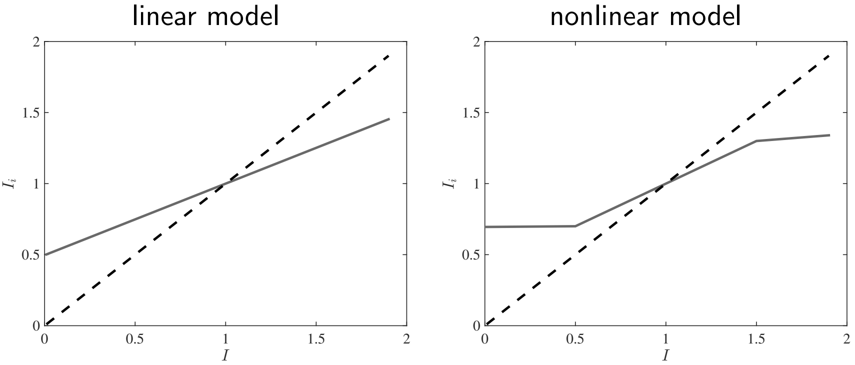
- $X_{it+1} = (1-\delta)X_{it} + I_{it}$
- $I_{it} = \alpha_0 - \alpha_1 X_{it} + \alpha_2 I_{it-1} + F\left(\frac{\sum I_{it}^s}{N}, u_t\right)$
- $F(I_t, u_t) = \widetilde{F}(I_t) + u_t.$
- $u_t = \rho u_{t-1} + \varepsilon_t$
- $\widetilde{F}$: S-shaped and piecewise linear

5. Quantitative Reduced Form Limit Cycle Model

Numerical example

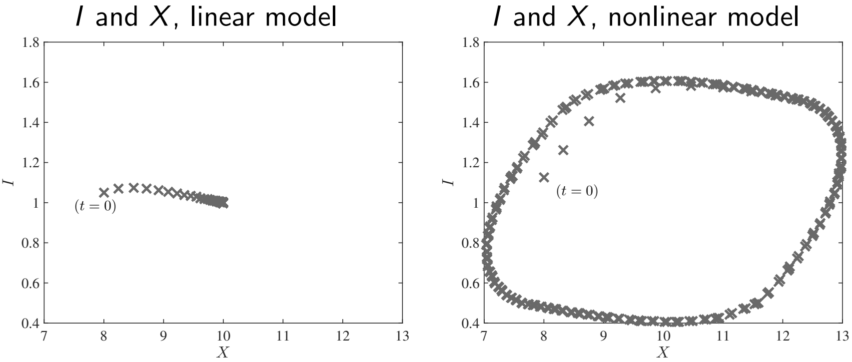
FIGURE 1 – Best response rules in the numerical example



- Steady state is $I_{SS} = 1$, $X_{SS} = 10$
- Deterministic simulation : let $X_0 = 8$, $I_0 = 1$

5. Quantitative Reduced Form Limit Cycle Model
Numerical example

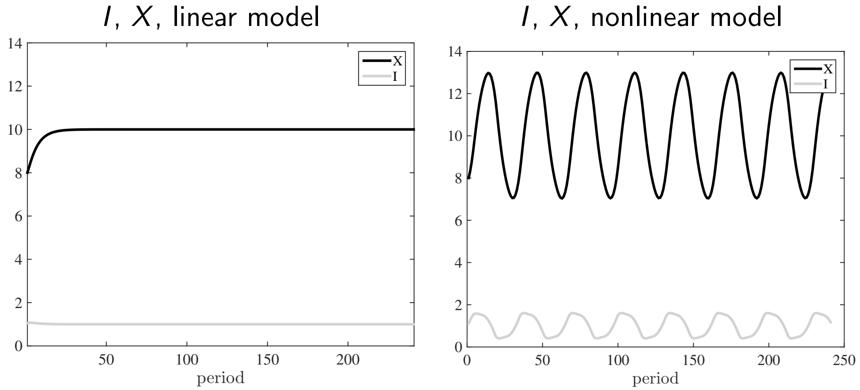
FIGURE 2 – Deterministic simulation



5. Quantitative Reduced Form Limit Cycle Model

Numerical example

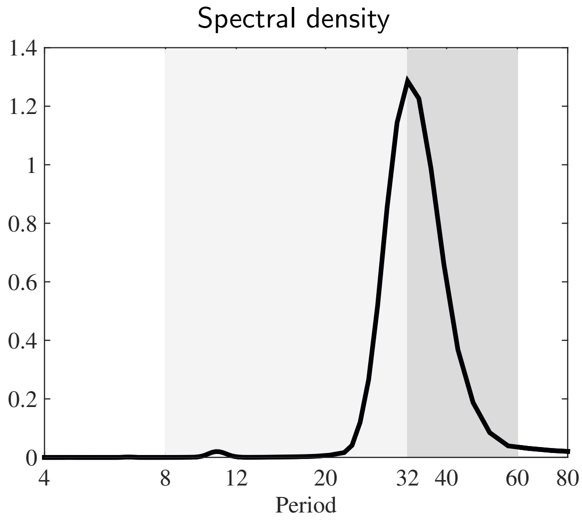
FIGURE 3 – Deterministic simulation



5. Quantitative Reduced Form Limit Cycle Model

Numerical example

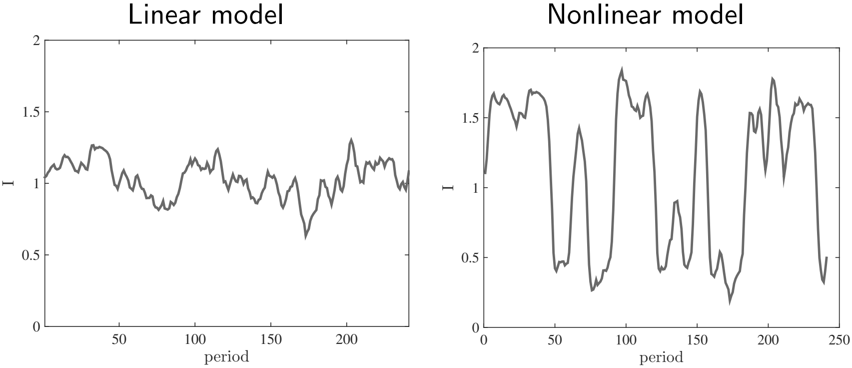
FIGURE 4 – Deterministic simulation of the nonlinear model



5. Quantitative Reduced Form Limit Cycle Model

Numerical example

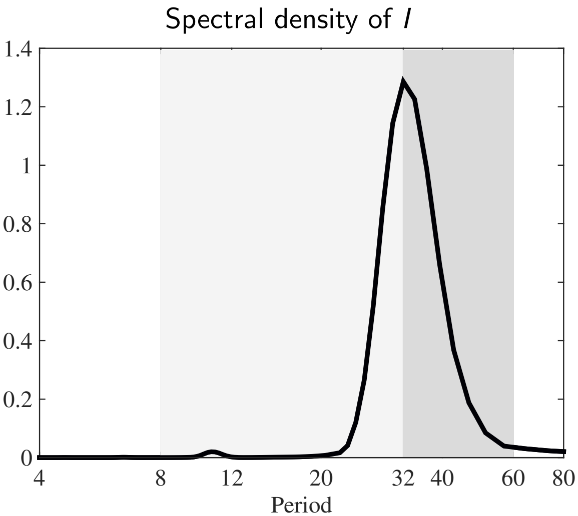
FIGURE 5 – One stochastic simulation



5. Quantitative Reduced Form Limit Cycle Model

Numerical example

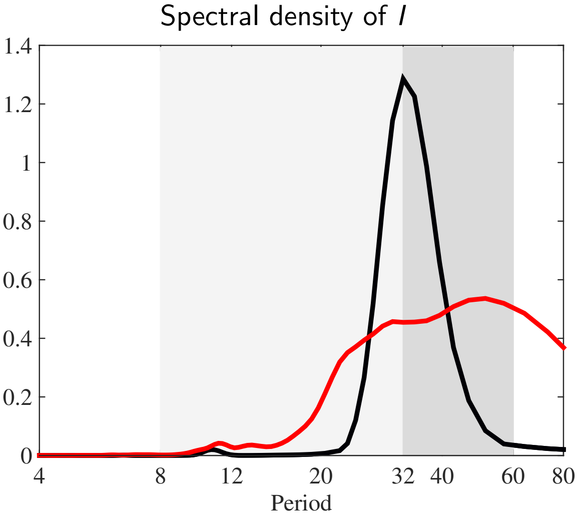
FIGURE 6 – Deterministic and Stochastic simulation of the nonlinear model



5. Quantitative Reduced Form Limit Cycle Model

Numerical example

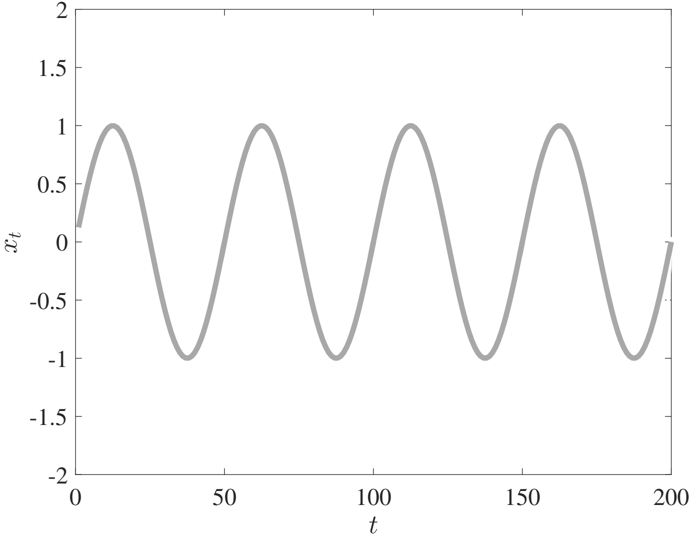
FIGURE 6 – Deterministic and Stochastic simulation of the nonlinear model



5. Quantitative Reduced Form Limit Cycle Model

What the results are not

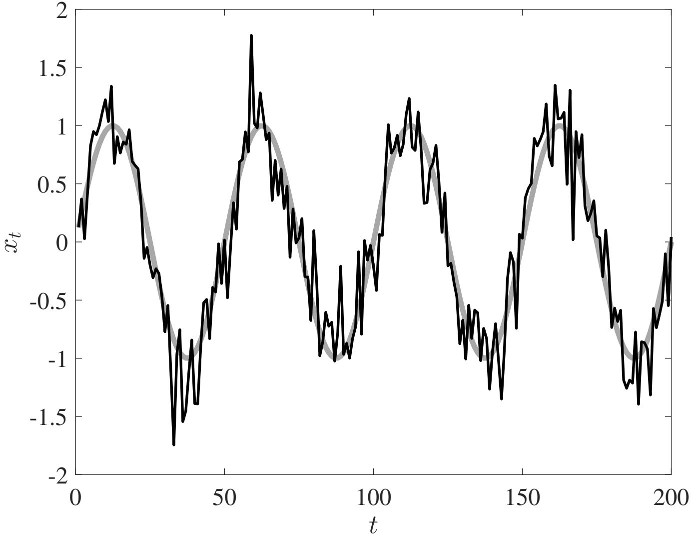
FIGURE 7 – $x_t = \sin(\omega t)$



5. Quantitative Reduced Form Limit Cycle Model

What the results are not

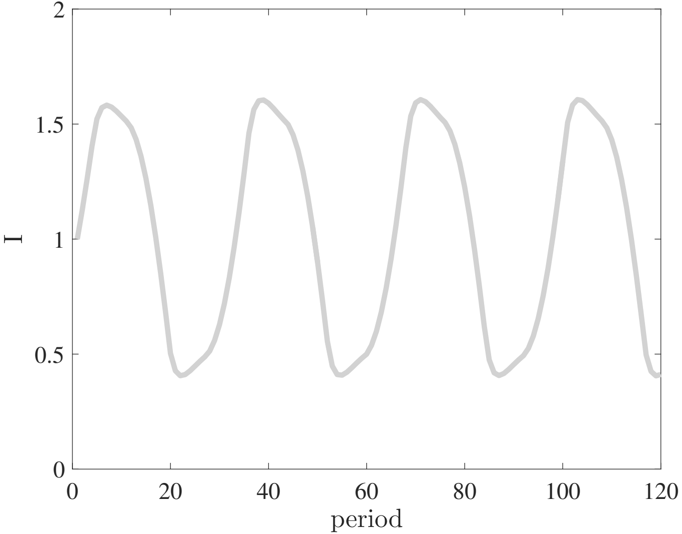
FIGURE 8 – $x_t = \sin(\omega t) + u_t$



5. Quantitative Reduced Form Limit Cycle Model

What the results are

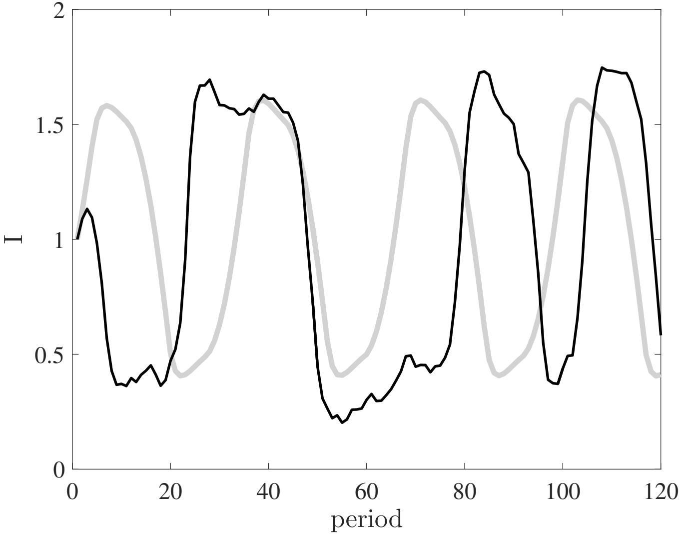
FIGURE 9 – Reduced Form Model



5. Quantitative Reduced Form Limit Cycle Model

What the results are

FIGURE 10 – Reduced Form Model



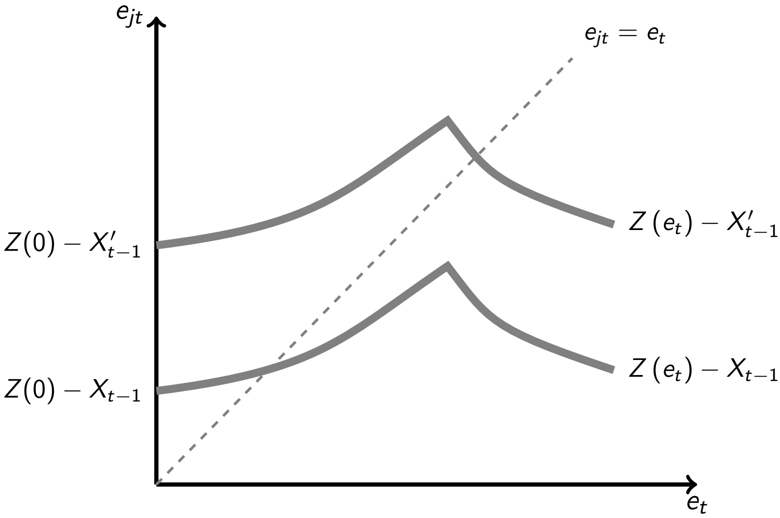
6. The Structural Model

Preferences and technology

- $$\sum_{t=0}^{\infty} \beta^t \left[\underbrace{U(c_{jt}) - \nu(\ell_{jt})}_{morning} + \underbrace{\widetilde{U}(\widetilde{c}_{jt}) - \widetilde{\nu}(\widetilde{\ell}_{jt})}_{afternoon} \right]$$
- Morning (Durable)
  - $c_{jt} = X_{jt} + e_{jt}$
  - $X_{jt+1} = (1-\delta)(X_{jt} + \gamma e_{jt})$
  - Matching : $M_t = M(N_t, L)$
  - Production in a match $\Theta_t F(\ell_t)$
  - N_t vacancies posted at cost Φ
  - A surplus sharing mechanism within a match
- Afternoon (Service)
  - Centralized labor and good markets
  - Total production $\widetilde{\Theta}_t F(\widetilde{\ell}_t)$
- IOUs between morning and afternoon

6. The Structural Model

FIGURE 11 – Equilibrium Determination - The case $\beta = 0$, "min" matching function and "competitive" wage setting



6. The Structural Model

What we aim to explain

- Let's estimate the model
- Does estimation select a configuration with limit cycles ?
- If so are the cycles sufficiently irregular and whether this is achieved with small noise.
- Our target is the behavior of hours : hours worked spectral density

6. The Structural Model

Target

FIGURE 12 – Hours, Level and Trend (High Pass Filter, 80 quarters)

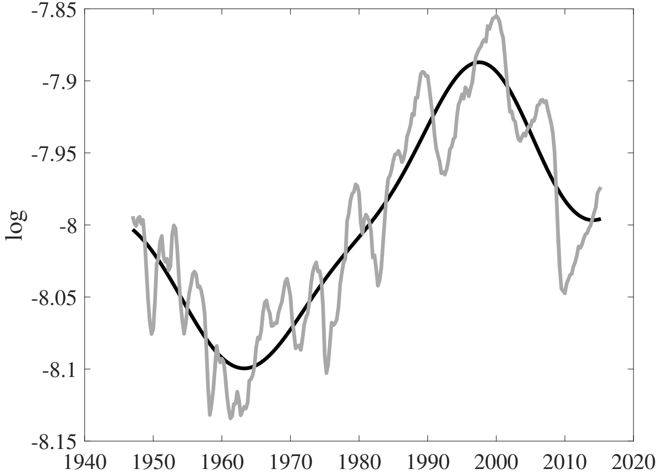


FIGURE 13 – Hours, Level and Cubic Trend

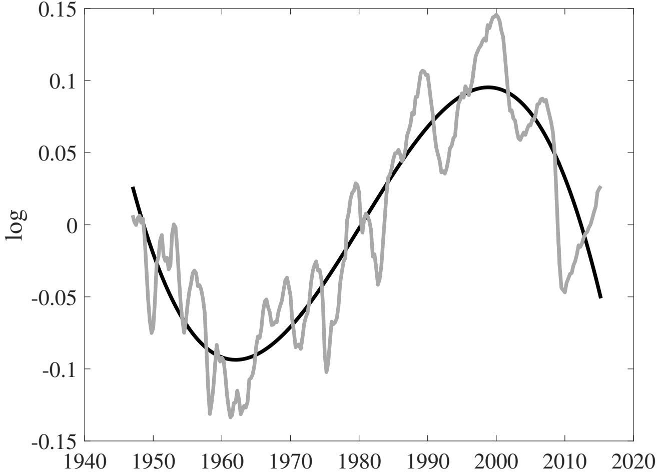
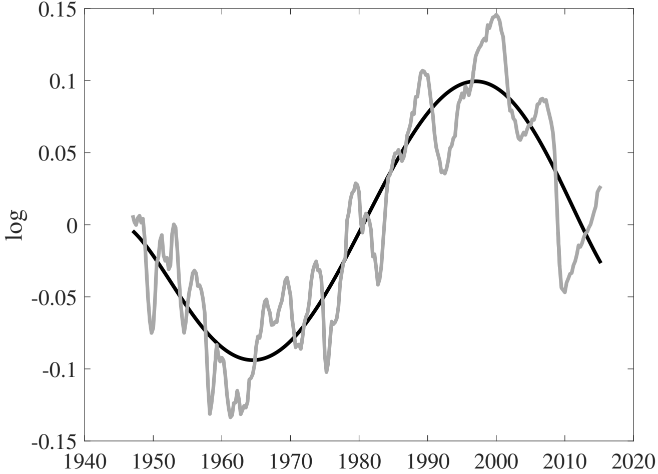


FIGURE 14 – Hours, Level and order 5 Trend



6. The Structural Model

Target

Figure 15 – Hours, Spectral Density

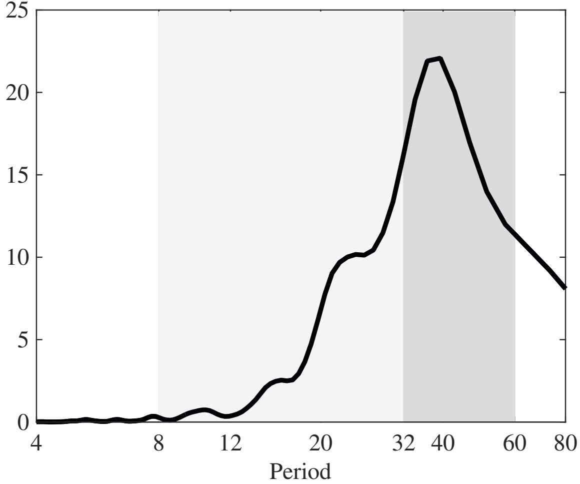
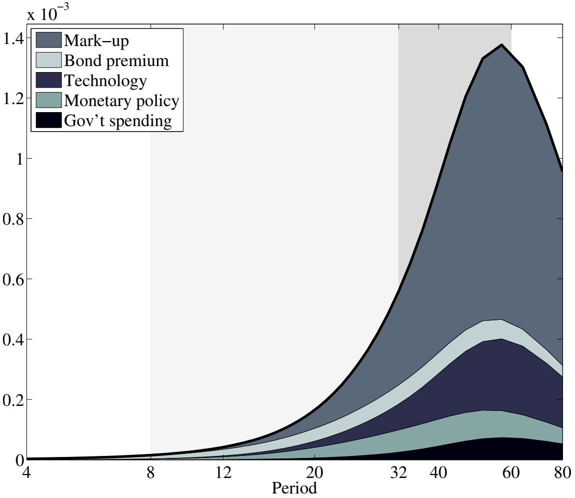


Figure 16 – Smets & Wouters (2007)



- Solution method : parameterized expectations
- Minimun distance estimation
  1. Simulate data from the model.
  2. Obtain spectrum for model from simulated data.
  3. Compute average squared difference from data spectrum Δ^2
- Find parameters that minimize Δ^2

FIGURE 17 – Hours, Estimated Spectral Density

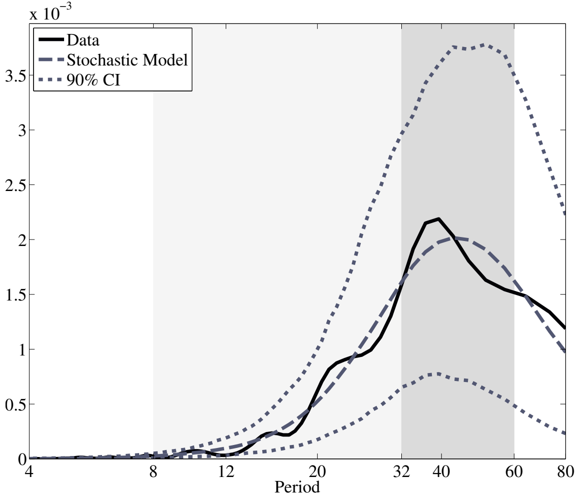
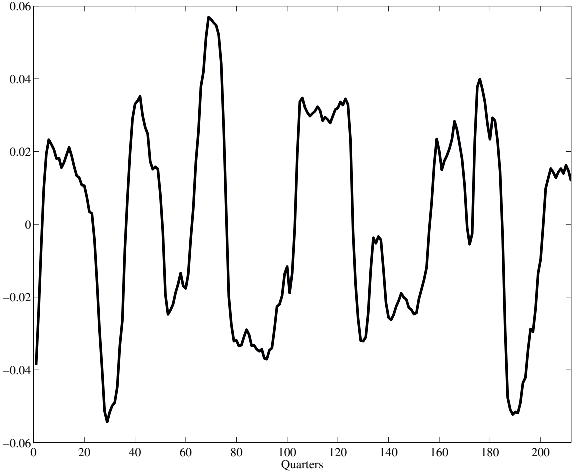


FIGURE 18 – Sample of Simulated Hours



- Let's simulate the model with estimated parameters but no shocks

FIGURE 19 – Hours, With No Shocks

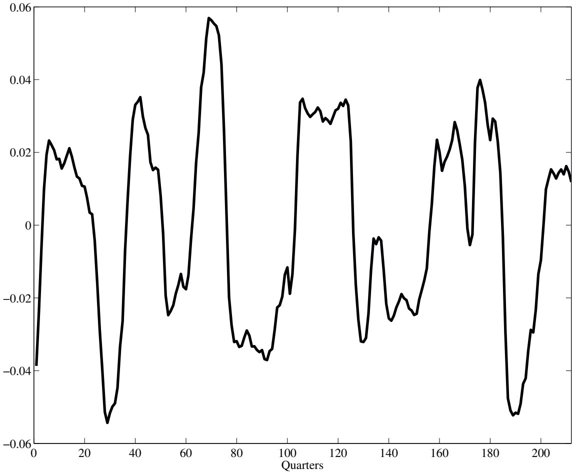
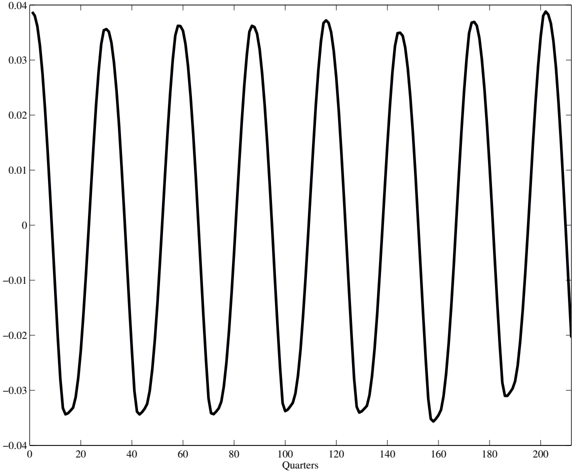


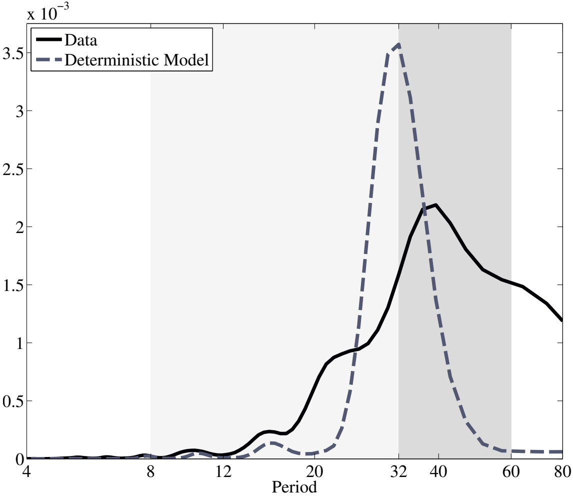
FIGURE 19 – Hours, With No Shocks



6. The Structural Model

Results

FIGURE 20 – Hours Spectrum, With No Shocks



- Standard deviation of hours :
  - × Deterministic model : 0.026.
  - × Stochastic model : 0.033.
  - × $\Rightarrow$ TFP accounts for 21% of s.d. of hours.
- If only shock in SMETS & WOUTERS (2007), s.d. $= 0.005$.
- Deterministic fluctuations models : parsimonious set of small shocks can generate sensible fluctuations.

FIGURE 21 – Hours Spectrum, Alternative Filters

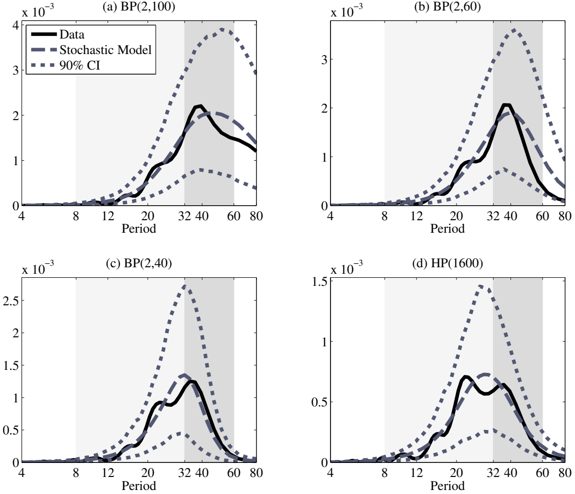
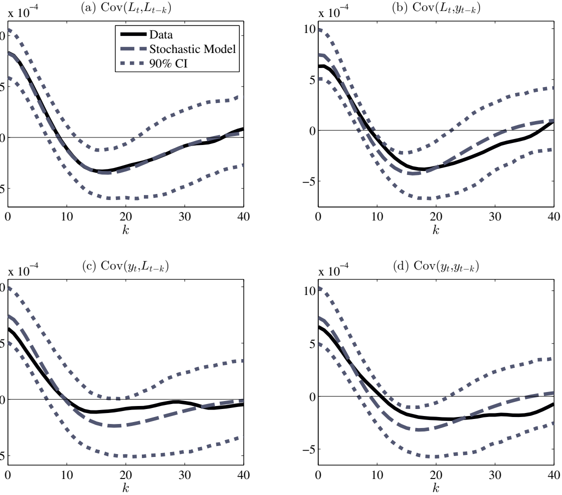


FIGURE 22 – Autocovariance : Hours Worked (L) and Output (y)



6. The Structural Model

Results

FIGURE 23 – Spectrum : Output (Data and Stochastic Model)